

GeNeSIS I-V: Architectural Unification in an Artificial-Life Simulation

An engineering case study from GeNeSIS I through V

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Abstract

GeNeSIS is a five-iteration artificial-life simulation developed as a single-author engineering project. The first four iterations explored, respectively, causal-emergence-based self-modeling, policy-gradient learning with inherited recurrent state, world-model-based planning, and a Hilbert-space representation for agent cognition. GeNeSIS V retains the fourth iteration's 64-dimensional complex state and uses it as the common substrate for cognition, genotype, and behavioral encoding. The design follows from a simple dimensional identity: the standard genetic code contains $4^3 = 64$ codons, matching the dimension of the cognitive state. In GeNeSIS V, an agent's genotype is therefore derived from the eigenbasis of its Hamiltonian rather than stored as a separate genome. The system then couples this substrate to independently modeled physical, chemical, developmental, social, and cultural processes, using established mathematical mechanisms where applicable, including reaction-diffusion dynamics, Arrhenius kinetics, predator-prey equations, autocatalytic chemistry, synchronization, associative memory, and island-biogeography-inspired population structure. The implementation contains 5,045 lines across twelve Python modules and is designed for a 2 GB memory ceiling. Verification combines module-level invariants, conservation tests, integration tests, and end-to-end testing of the deployed application. Several implementation defects were found and corrected during verification. Two substantive negative findings remained: the current parameterization does not produce tribal diversification or phylogenetic diversification. These failures are reported as results rather than concealed. The paper presents GeNeSIS V as an architectural case study, not as evidence for machine consciousness or as a completed solution to open-ended artificial life.

Keywords: artificial life, multi-agent systems, computational biology, integrated information, open-ended evolution, quantum-inspired cognition

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1 Introduction

Artificial-life research has demonstrated repeatedly that complex behavior can arise from relatively simple local rules, selection, and interaction. Systems such as Tierra [18], Avida [17], and PolyWorld [26] established different approaches to digital evolution, self-replication, and embodied agents. Later work on open-endedness introduced mechanisms such as novelty search and environment-agent coevolution to reduce evolutionary stagnation [2, 13, 25]. Most such systems, however, specialize around a particular representation or learning mechanism.

GeNeSIS explores a different engineering question: what changes when several previously separate mechanisms are required to share a common mathematical representation? The project evolved through five successive implementations. Each iteration was built around the limitations of the preceding one, and the fifth iteration became the focus of the present report because it introduced a common substrate for cognition, genetics, and behavioral encoding.

The central design choice in GeNeSIS V is not that a cognitive state should resemble a genome, but that the two can be represented by the same mathematical object. The cognitive state has dimension 64, while the standard genetic code has 64 possible codons. GeNeSIS V uses this equality directly: the eigenbasis of an agent’s Hamiltonian provides the 64 positions from which a codon string can be derived. A change in the Hamiltonian is consequently a change in both cognition and genotype. This removes an entire synchronization relationship from the architecture.

The project is best classified as an engineering case study rather than a controlled scientific study. GeNeSIS is single-authored and has not yet undergone independent replication. The contribution claimed here is therefore narrow: the report documents a concrete architecture, its implementation constraints, its verification methodology, and both successful and unsuccessful behaviors observed under the specified parameters.

The remainder of the paper is organized as follows. Section 2 places the design in context. Section 3 summarizes GeNeSIS I-IV. Section 4 describes the GeNeSIS V architecture. Section 5 gives the mathematical methods. Section 6 discusses implementation constraints, Section 7 describes verification, Section 8 reports the observed results, Section 9 interprets those results, and Section 10 states the limitations.

2 Related Work

2.1 Artificial Life and Open-Endedness

Classical artificial-life systems showed that complex digital behavior can emerge from local rules operating under selection [17, 18, 26]. A recurring difficulty is evolutionary stagnation, in which populations converge on a narrow region of the search space. Novelty search addresses this by rewarding behavioral distinctiveness rather than task performance alone [13]. POET and minimal-criterion coevolution extend the idea by changing the environment as agents evolve [2, 25]. GeNeSIS V incorporates novelty-based reward shaping and population-level adaptation, but, as discussed later, does not yet implement full environment-agent coevolution.

2.2 Integrated Information as a Design Instrument

Integrated Information Theory (IIT) proposes a measure of informational integration denoted by Φ [21]. IIT remains one of several competing frameworks for understanding consciousness, and the presence of a Φ -like quantity in a simulation does not establish subjective experience. GeNeSIS V therefore uses an explicitly defined integration proxy only as an internal state metric. The purpose is operational: to quantify how the represented state changes under a chosen partition. No metaphysical interpretation is required for the metric to be useful as an engineering

diagnostic.

2.3 Quantum-Inspired Cognitive Modeling

Quantum-probability formalisms have been used in mathematical models of human judgment without requiring a claim that human cognition is physically quantum-mechanical [3]. GeNeSIS IV and V adopt the same modeling stance. Complex Hilbert-space states, unitary evolution, and Born-rule action selection are used because they provide a compact representation of alternatives and allow interference-like structure in the state dynamics. The implementation is classical software; no physical quantum process is assumed.

2.4 Mechanistic Submodels

Where a subsystem can be grounded in an established quantitative model, GeNeSIS V uses that model rather than introducing an arbitrary game mechanic. The implementation draws on Turing’s reaction-diffusion framework [22], the Gray-Scott reaction-diffusion system [6], Arrhenius kinetics [1], Swadesh glottochronology [20], Lotka-Volterra dynamics [15, 24], autocatalytic-set theory [9, 10], sexual-selection models [4, 12, 27], island biogeography [16], the Red Queen hypothesis [23], Kuramoto synchronization [11], Hopfield associative memory [8], Godel numbering [5], and Lindenmayer systems [14].

3 Project Lineage: GeNeSIS I-IV

The first four iterations were completed before the present study. They are summarized for architectural context; their implementations were not independently re-verified as part of the GeNeSIS V verification campaign.

3.1 GeNeSIS I: Causal-Emergence-Based Self-Modeling

The first implementation combined an Advantage Actor-Critic (A2C) agent with a measurement layer intended to capture homeostatic regulation, predictive self-modeling, and a hierarchy of behavioral milestones. Causal-emergence and free-energy ideas were used as design vocabulary. The original documentation made stronger claims about simulated consciousness than are warranted by the present methodology; those claims are not repeated here.

3.2 GeNeSIS II: Policy-Gradient Learning with Inherited Recurrent State

The second iteration replaced A2C with Proximal Policy Optimization (PPO) [19] operating on a GRU policy over a 41-dimensional observation space. A newly spawned agent inherited a blend of its parents’ terminal recurrent states rather than starting from a fresh hidden state. This was intended as a computational analogue of inherited memory. Population dynamics included birth, death, and a meme-pool mechanism for horizontal transfer between agents.

3.3 GeNeSIS III: World-Model-Based Planning

The third iteration adopted a Dreamer-style recurrent state-space model [7]. Agents could evaluate short imagined trajectories before selecting an action, separating action choice from immediate sensory state. The iteration also introduced interface components for emergent milestones and cultural visualization. Those design elements informed the later Biome Cartography interface in GeNeSIS V.

3.4 GeNeSIS IV: Hilbert-Space Cognitive Architecture

GeNeSIS IV replaced the recurrent-network core with a complex state vector $\psi \in \mathbb{C}^{64}$. Each agent evolved this state under a personal Hermitian Hamiltonian, selected actions through Born-rule probabilities, and learned through Hamiltonian perturbations. An IIT-inspired integration measure, Godel-numbered behavioral programs, a Hermitian social-memory layer, and Kuramoto synchronization were also introduced. The 64-dimensional state established the mathematical substrate later reused in GeNeSIS V.

4 GeNeSIS V Architecture

4.1 The Central Construction

The GeNeSIS IV cognitive state has dimension $K = 64$. The standard genetic code contains $4^3 = 64$ codons. GeNeSIS V uses this identity as an architectural constraint rather than as a biological claim.

Let H be an agent’s Hermitian Hamiltonian with eigenvectors v_k , $k = 1, \dots, 64$. For each eigenvector, the phase of its dominant component is mapped to one of four symbolic nucleotide bases by phase quadrant. The resulting 64-character string is interpreted as a codon sequence under the standard genetic code. In this representation, genome and cognition are not stored independently. A perturbation of H changes the cognitive dynamics and also changes the derived genotype.

The important consequence is architectural rather than biological: the implementation no longer has to maintain consistency between two independently mutable state objects. Learning updates, inherited variation, and Hebbian changes all operate on the same Hamiltonian representation.

4.2 System Overview

GeNeSIS V consists of twelve Python modules totaling approximately 5,045 lines, together with a Streamlit front end. The modules are grouped into six functional domains and a cross-cutting evaluation layer.

Module	Role
<code>metacognition.py</code>	Godel encoding, associative memory, novelty scoring, phylogenetic clustering
<code>consciousness.py</code>	Hilbert-space state evolution, Born-rule action selection, Φ proxy
<code>biology.py</code>	Codon reader and Gray-Scott morphogenesis
<code>world.py</code>	Procedural terrain, resources, stigmergic fields, frozen physics oracle
<code>agents.py</code>	Embodied agent lifecycle, reproduction, and death
<code>evolution.py</code>	Population bounds, meta-fitness, cultural-transmission checks
<code>chemistry.py</code>	Elemental composition, reaction kinetics, molecular speciation
<code>civilization.py</code>	Tribal formation, diplomacy, and technology graph
<code>geometry.py</code>	Procedural growth grammars and topological invariants
<code>narrative.py</code>	Oral-tradition transmission model
<code>nobel.py</code>	Six-category breakthrough detection
<code>GeNeSIS_V.py</code>	Interactive Streamlit front end

Table 1: GeNeSIS V module structure.

5 Methods

5.1 Cognitive Dynamics

Each agent has a normalized state $\psi \in \mathbb{C}^{64}$ with $\|\psi\| = 1$. If

$$H = V \text{diag}(\lambda) V^\dagger$$

is the eigendecomposition of the Hermitian Hamiltonian, the state update is

$$\psi(t + \Delta t) = V \text{diag} \left(e^{-i\lambda\Delta t} \right) V^\dagger \psi(t). \quad (1)$$

Candidate action states are constructed from the eigenbasis and modulated by a Fourier encoding of the observation. Action probabilities are proportional to

$$P(a_i) \propto |\langle b_i | \psi \rangle|^2,$$

with a curiosity-dependent temperature controlling sampling sharpness.

Learning perturbs H with a reward-signed, meta-modulated rank-one Hermitian update. The implementation assigns a cognitive cost proportional to the absolute change in von Neumann entropy,

$$c = \kappa |\Delta S|,$$

where S is computed in the current eigenbasis. This is used as a computational analogue of information-processing cost and is not a thermodynamic measurement of the software.

5.2 Integrated-Information Proxy

The state is partitioned into a 24-dimensional task subspace and a 40-dimensional complement. The implementation defines

$$\Phi = I(\psi_{\text{task}}) + I(\psi_{\text{comp}}) - I(\psi), \quad (2)$$

where

$$I(x) = \log_2 (1 + \text{Var}(|x|)),$$

and negative values are clipped at zero. Sustained increases are flagged when the trailing ten-step mean exceeds 1.2 times the mean measured 40-50 steps earlier. This is an internal consistency metric, not a validated measure of consciousness.

5.3 Morphogenesis

Biome patterning, and prospectively developmental body-plan patterning, use the Gray-Scott reaction-diffusion equations [6]:

$$\partial_t A = D_A \nabla^2 A - AB^2 + f(1 - A), \quad (3)$$

$$\partial_t B = D_B \nabla^2 B + AB^2 - (f + k)B. \quad (4)$$

The equations are integrated explicitly on a toroidal grid using a five-point discrete Laplacian. The same discrete operator is reused for environmental heat diffusion.

5.4 Reaction Kinetics

Chemical reactions are represented over molecular species rather than only bulk elemental composition. Reaction rates follow the Arrhenius form

$$k(T) = A \exp\left(-\frac{E_a}{RT}\right), \quad (5)$$

with conversion gated by the rate-limiting reactant at each simulation step.

5.5 Cultural Transmission

A tribe’s founding myth is represented as a sequence of 16 symbols drawn from the same primitive behavioral vocabulary used in Godel encoding. Transmission uses a Swadesh-style retention model [20]: each symbol is retained independently with probability r per generation. The nominal value is $r = 0.805$, increased toward 0.955 in the presence of a local sacred stigmergic signal. Cultural fidelity is measured as Pearson correlation between the founding sequence and the current sequence.

5.6 Social Structure

Tribal power combines eight quantities: aggregate energy, aggregate health, discovery count, membership, alliance count, mean metacognitive spread, mean Φ , and trade volume. Alliances dissolve under an epistemic-schism rule when the Euclidean distance between the mean meta-Hamiltonian eigenspectra of two tribes exceeds a threshold Θ_{schism} .

Reproduction blends parental ψ and H states and then applies independent Hermitian perturbations. The parental contribution factor is sampled as $\alpha \sim U(0.35, 0.65)$.

5.7 Open-Endedness Mechanism

The design uses novelty-based reward shaping, with behavioral programs compared through a Godel-derived distance to previously observed programs. This provides an agent-side novelty mechanism. It does not, however, implement the environment-agent coevolution component represented by POET and minimal-criterion coevolution [2, 25]. The claim of open-endedness is therefore intentionally limited to the mechanisms actually implemented.

6 Implementation and Engineering Constraints

The entire system was designed around a 2 GB memory ceiling representative of low-cost consumer or free-tier cloud hardware. Memory control was treated as an architectural requirement rather than a final optimization pass.

Terrain is procedural and evaluated on demand rather than materialized in full. Each agent’s full-precision cognitive state occupies approximately 90 KB, which remains modest at the configured 128-agent ceiling. Historical data are kept in bounded ring buffers rather than unbounded logs. Dead agents are removed immediately after their terminal state has been distributed to neighboring agents within the same step, preventing population history from becoming a hidden memory leak.

The PyTorch dependency used in the GeNeSIS IV design was removed in V for the environment’s frozen physics oracle. A numerically equivalent NumPy multilayer perceptron was used instead, avoiding a large baseline import cost for a component that is not trained at runtime.

7 Verification Methodology

Verification was performed at three levels: module invariants, cross-module integration, and end-to-end application behavior.

Each module contains a direct test that checks behavior-specific invariants rather than merely verifying import success. The cognitive core was tested for preservation of $\|\psi\| = 1$ and Hermiticity of H over tens of thousands of updates. Chemical reactions were tested for atomic-mass conservation. Topological routines were checked against synthetic cases with known Betti numbers: a solid square with one enclosed hole gives $\beta_0 = 1, \beta_1 = 1$, while two disconnected regions give $\beta_0 = 2$. A regression routine used by the physics oracle was first validated against an exact linear function before being applied to the nonlinear environment.

Cross-module tests executed full agent populations through the cognition-action-learning-evolution pipeline. The deployed Streamlit application was also tested with Streamlit’s AppTest framework, using the actual application script rather than a mock. Runs of up to 200 steps were used to verify that all seven interactive panels executed against accumulated live state without exceptions.

8 Results

The results below include both corrected implementation errors and behaviors that remained unresolved under the specified parameters.

8.1 Population Regulation

When population bounds were disabled during preliminary isolated tests, the embodied-agent module produced substantial run-to-run variation. Across 300 steps, observed surviving populations ranged from 5 to 130 agents under nominally similar conditions. The population-bounds subsystem was then enabled, using meta-fitness-weighted immigration below a floor of 28 agents and culling above a ceiling of 128. In an independent 400-step verification run, the observed population remained in the interval $[29, 70]$, confirming that the regulation mechanism behaved as designed.

8.2 Representational Error in Chemical Kinetics

The first implementation represented reactions directly on bulk elemental composition. That representation cannot produce a molecular conversion effect for a mass-balanced reaction because reactant and product elemental totals are identical by construction. The problem was detected by testing a concrete molecular consequence: whether glucose concentration fell under repeated respiration events.

The implementation was revised to track molecular species separately from bulk elemental totals. After correction, glucose concentration decreased from 0.114 to 0.030 arbitrary concentration units over 20 steps while total elemental mass remained conserved.

8.3 Miscalibration in the Evaluation Layer

The initial six-category breakthrough detector produced 33 detections during a 30-step live run, which was inconsistent with its intended role of identifying rare events. Two causes were isolated. First, the locally fitted polynomial causal model generalized too far because the environment’s spatial fields are smooth over much of the domain. The evaluation distance was therefore extended from 4-7 cells to 12-20 cells. Second, an economic-efficiency detector accepted any

positive improvement over the previous record; it was changed to require at least a 15% relative improvement plus a minimum recurrence interval.

After both corrections, an independent 200-step run produced seven breakthrough detections across two of the six categories.

8.4 Asymmetric Statistical Comparison

A lineage-longevity detector was intended to identify a lineage whose mean lifespan exceeded the population baseline. The original implementation included the candidate lineage in that baseline, allowing extreme values from the lineage under test to raise the threshold used to judge the same lineage. A constructed 50-versus-90-step comparison failed to trigger the detector.

The baseline was corrected to a leave-one-group-out estimate. The same constructed case then triggered as expected.

8.5 Negative Finding: No Tribal or Phylogenetic Diversification

Under the fully specified GeNeSIS V parameters, no test run produced more than one tribe. This included a 500-step run initialized with 32 agents. Alliance, war, and schism mechanisms therefore did not activate in an integrated simulation.

The failure was traced to the spectral-resonance similarity measure used for tribal affiliation. Across independently initialized pairs at birth, resonance values ranged from 0.986 to 0.9996, while the affiliation threshold was 0.08. In practice, every pair passed the threshold, preventing the system from founding a second tribe.

The resonance calculation itself matched its specification. The underlying structural cause was traced to Hamiltonian construction: every agent’s diagonal was generated from the same ordered sequence of prime numbers and differed only in amplitude. This produces highly similar eigenvalue shapes across the population despite differences in individual history.

Phylogenetic clustering showed an analogous failure and never produced more than one clade. Because these observations follow directly from the specified parameterization and not from an implementation defect, the constants were not changed retrospectively to force diversification.

9 Discussion

The principal result is narrow but concrete. GeNeSIS V demonstrates an implementable architecture in which cognition, genotype, and behavioral encoding are derived from one 64-dimensional mathematical substrate. The verification results indicate that this unification does not, by itself, invalidate the local correctness checks applied to the constituent modules.

The same result should not be read more broadly than the evidence permits. The presence of a Hilbert-space state, an IIT-inspired integration proxy, or Godel-encoded behavioral programs does not establish understanding, intention, agency, or subjective experience. These components are computational constructions selected for specific engineering purposes.

The diversification failure is also informative. A system that reports only successful mechanisms can obscure the interaction effects that become visible only in an integrated implementation. Here, the social and phylogenetic layers are present and locally testable, yet remain inactive under the global parameterization because the population is too spectrally homogeneous at initialization. This is a meaningful architectural limitation and a direct guide for the next iteration.

10 Limitations

1. **Single-author development.** The codebase has not received independent review or replication, and its design reflects the decisions of a single author.
2. **No tribal or phylogenetic diversification.** The current parameterization produces a single tribe and a single phylogenetic clade in integrated runs. Social and speciation mechanisms were unit-tested but were not observed to activate at the population level.
3. **Limited time horizon.** Verification covered isolated runs of up to approximately 500 steps and full application runs of up to 200 steps. Behavior over thousands of steps remains untested.
4. **Author-calibrated evaluation thresholds.** The breakthrough detector was calibrated against a small number of test runs and has no external benchmark for scientific or cultural significance. Its category labels are descriptive engineering labels, not equivalents of formal awards or scientific institutions.
5. **No evidence for consciousness.** The reported Φ proxy and Hilbert-space formalism are computational devices and provide no evidence of subjective experience.
6. **Prior iterations were not re-verified.** GeNeSIS I-IV are summarized from their design documentation rather than re-tested in this study.
7. **Partial open-endedness support.** Novelty-based reward shaping is implemented, but environment-agent coevolution is not. Long-term resistance to evolutionary stagnation is therefore not established.

11 Conclusion

GeNeSIS V provides a concrete example of architectural unification in an artificial-life simulation. A single 64-dimensional complex substrate is used to support cognitive state evolution, genotype derivation, and behavioral encoding, after which independently modeled physical, chemical, developmental, social, and cultural processes are coupled to the resulting agents. The implementation was engineered under a 2 GB memory ceiling and verified through invariant checks, conservation tests, integration tests, and end-to-end application tests.

The work also demonstrates the importance of reporting failures. Two mechanisms that were explicitly implemented - tribal diversification and phylogenetic diversification - did not occur under the current parameterization. Rather than adjusting those parameters retrospectively, the report traces the behavior to the spectral structure imposed on agent Hamiltonians and records it as an unresolved limitation.

The appropriate interpretation of GeNeSIS is therefore methodological. It is an engineering case study in how a heterogeneous artificial-life system can be constructed, verified, constrained, and diagnosed when multiple subsystems are required to share a common representation. It is not a demonstration of machine consciousness, and it does not claim to solve open-ended artificial life.

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